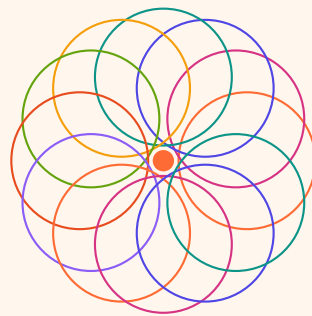
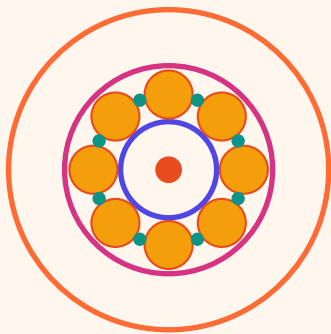


# ■ MATHS ART

## MANDALA SPIROGRAPH

Integrated Lesson Plan



|          |                                |
|----------|--------------------------------|
| Subject  | Mathematics + Art (Integrated) |
| Duration | 45–60 minutes                  |
| Theme    | Circles, Patterns & Mandalas   |

## ■ LESSON OVERVIEW

This integrated Maths-Art session introduces students to fundamental geometric concepts — **circles, centre points, and repeating patterns** — through the culturally rich lens of Indian **Mandala** art. Students engage in hands-on creation of their own mandala using everyday household objects, developing both mathematical thinking and artistic expression simultaneously.

### Big Idea

Mathematics is not only numbers and equations — it lives in art, culture, and beauty. Circles, symmetry, and patterns are the building blocks of mandalas found in Indian temples, Rangoli designs, and Kolam art. When children create a mandala, they ARE doing real mathematics.

### Learning Objectives

| Domain        | Students Will Be Able To...                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Knowledge     | <ul style="list-style-type: none"><li>• Define circle, centre, and pattern in their own words</li><li>• Identify circles in their everyday environment</li><li>• Explain that 'Mandala' means 'circle' in Sanskrit</li></ul>                    |
| Skills        | <ul style="list-style-type: none"><li>• Trace circles of different sizes using round objects</li><li>• Draw repeating petal and dot patterns around a central point</li><li>• Create a colour pattern using at least 2 colours</li></ul>        |
| Understanding | <ul style="list-style-type: none"><li>• Understand that patterns are shapes/colours that repeat</li><li>• Recognise symmetry by observing their finished mandala</li><li>• Connect mathematical concepts to Indian cultural art forms</li></ul> |
| Attitude      | <ul style="list-style-type: none"><li>• Appreciate mathematics as creative and beautiful</li><li>• Feel proud of their own artistic-mathematical creation</li></ul>                                                                             |

## ■ CURRICULUM CONNECTIONS & MATERIALS

### Curriculum Links

| Curriculum Area        | Specific Concept           | Connection to Activity                                 |
|------------------------|----------------------------|--------------------------------------------------------|
| Mathematics (Geometry) | 2D shapes: circle          | Students trace and identify circles of different sizes |
| Mathematics (Patterns) | Repeating patterns         | Colour and shape patterns around the mandala ring      |
| Mathematics (Spatial)  | Centre point, symmetry     | Starting from centre; equal petals all around          |
| Art & Craft            | Drawing, colouring, design | Creating a personal mandala artwork                    |
| EVS / Social Studies   | Indian culture & heritage  | Rangoli, Kolam, Madhubani, temple art context          |
| Language               | Vocabulary acquisition     | Learning: circle, centre, pattern, mandala, repeat     |

### Materials Required

| Material              | Purpose                                | Qty / Student |
|-----------------------|----------------------------------------|---------------|
| Plain white A4 paper  | Drawing surface for the mandala        | 2 sheets      |
| Pencil                | Drawing centre dot and initial circles | 1             |
| Coin / small bangle   | Tracing the inner circle               | 1             |
| Small bowl or lid     | Tracing the middle circle              | 1             |
| Large bowl / plate    | Tracing the outer circle               | 1 (shared)    |
| Sketch pens / crayons | Colouring with patterns                | 4+ colours    |
| Black sketch pen      | Outlining petals and dots              | 1             |

## ■ KEY VOCABULARY

Introduce these five words explicitly. Use gestures, choral repetition, and bilingual connections to cement understanding.

### ■ CIRCLE

A perfectly round 2D shape with no corners, no edges. Every point on a circle is the same distance from the centre.

■■■ **Teacher tip:** Hold up a coin. Trace it in the air with your finger.

### ■ CENTRE

The exact middle point of a circle. All other parts of a mandala grow outward from this one point.

■■■ **Teacher tip:** Point to your belly-button — that's YOUR centre!

### ■ PATTERN

Shapes or colours that repeat in a predictable order. Patterns are everywhere — on sarees, tiles, and nature.

■■■ **Teacher tip:** Clap-clap-stamp, clap-clap-stamp! Body patterns count too.

### ■ MANDALA

A circular design that radiates outward from a centre point. Found in temples, Rangoli, and Kolam art across India.

■■■ **Teacher tip:** Show examples: Diwali Rangoli, temple ceiling, Kolam.

### ■ REPEAT

To do or show something again and again in the same way. Repeating a shape or colour creates pattern and symmetry.

■■■ **Teacher tip:** Say it together: repeat, repeat, REPEAT!

## LESSON FLOW & TIMING

| Phase                     | Activity                                                                                                                      | Duration | Key Questions / Actions                                                                          |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------|
| ■ Hook & Wonder           | Show images of Rangoli, Mandala, and Kolam. Ask: "Have you seen this at home? Where?"                                         | 5 min    | What shapes do you see? What is in the middle?                                                   |
| ■ Concept: Circle         | Discuss what a circle is. Students identify circles in the classroom (clock, plate, coin).                                    | 5 min    | Can circles have corners? Are all circles the same size?                                         |
| ■ Concept: Centre         | Demonstrate finding the centre of a circle drawn on the board. Body movement: point to own belly button.                      | 3 min    | Where would you put the first dot on your paper?                                                 |
| ■ Concept: Pattern        | Clap-clap-stamp body pattern. Show examples: saree border, Rangoli, Madhubani. Students continue patterns on mini whiteboard. | 5 min    | What comes next? How do you know?                                                                |
| ■ Concept: Mandala        | Show mandala examples: Diwali Rangoli, temple ceiling, Kolam. Link: circle + pattern + centre = Mandala!                      | 4 min    | What do all these designs have in common?                                                        |
| ■ Vocabulary Time         | Introduce 5 words with gestures and choral repetition. Students say each word in English and local language.                  | 3 min    | Repeat after me: CIRCLE! CENTRE! PATTERN! MANDALA! REPEAT!                                       |
| ■ Create Time (Steps 1–6) | Students follow 6-step guided process to create their own mandala using coins, bowls, pencil, sketch pens.                    | 18 min   | Check: Is your centre dot in the middle? Are your petals repeating? What is your colour pattern? |
| ■ Share & Celebrate       | Gallery walk or show-and-tell. Teacher asks 3 recall questions. Students earn 'Mandala Maths Artist' title.                   | 5 min    | What shape is a mandala? What is the middle called? When something repeats, it's a...?           |
| ■ Preview Next Session    | Tease Spirograph session: 'More circles, more patterns, more magic!'                                                          | 2 min    | What do you think a Spirograph might look like?                                                  |

## ⇒ ■ STEP-BY-STEP: CREATING YOUR MANDALA

Follow these 6 steps in order. Each step builds on the previous one. Encourage students to go slowly and carefully — this is maths AND art!

### THE CENTRE DOT ■

Place your paper on a flat surface. Find the middle of the page — this is where your mandala will live. Make a small, clear dot right in the centre with your pencil. **This is the most important step** — everything else grows from this point!

■■■ **Teacher Note:** Ask: 'If the centre is in the wrong place, what happens to the whole mandala?' Mathematical link: the centre is equidistant from all points on the circle's edge.

### 2 TRACE THE FIRST (SMALL) CIRCLE ■

Place a coin flat on the paper, positioning it exactly over your centre dot. Hold the coin steady with one finger, then draw carefully around its edge with your pencil. Lift the coin — you have your first circle! **This is your inner ring.**

■■■ **Teacher Note:** Demonstrate slowly. Emphasise: hold the object still, move the pencil. Mathematical link: the coin is a physical representation of a circle — all points equal distance from centre.

### 3 TRACE A BIGGER CIRCLE ■

Now use a larger round object — a small bowl or a bottle cap. Place it so its centre lines up with your dot. Trace around it. This creates your **middle ring**. Watch the space between the two circles — like ripples spreading out from a stone in water!

■■■ **Teacher Note:** The concentric circles reinforce the concept that circles share a centre. Mathematical vocabulary: concentric circles.

### 4 ADD PETALS ■

Draw petal shapes in the space between your two circles — like a sunflower's petals! Each petal should touch the inner circle at the bottom and the outer circle at the top. **Make the same number of petals all the way around.** Make sure each petal repeats the same shape — that's your pattern!

■■■ **Teacher Note:** Tip: Lightly pencil-mark equal spacing first (fold paper to find quarters). Mathematical link: equal spacing, rotational symmetry.

## 5

## ADD DOTS OR LINES ■

Look at the spaces between each petal. In each space, draw a small dot or a short line. These tiny details make your mandala sparkle — just like Diwali Rangoli dots! **Repeat the same decoration in every space** to keep the pattern.

■■■ **Teacher Note:** Connect to Rangoli tradition — dots and lines create the texture of traditional Indian art. Mathematical link: alternating pattern in the gaps = ABAB pattern.

## 6

## COLOUR WITH A PATTERN ■

Choose 2 or more colours. Decide on a colour pattern — for example: yellow, orange, yellow, orange... OR red, blue, red, blue. **Apply the same pattern all the way around your mandala.** Don't forget to colour the outer ring too!

■■■ **Teacher Note:** Ask students to state their colour pattern out loud before colouring. Mathematical link: ABAB, ABCABC, or AABB repeating colour patterns.

## ■ ASSESSMENT & DIFFERENTIATION

### Formative Assessment

Assessment in this session is primarily observational and dialogic. Use the questions below during the Create Time and the Share & Celebrate phases.

| Assessment Focus       | Questions to Ask                                                           | Look For                                            |
|------------------------|----------------------------------------------------------------------------|-----------------------------------------------------|
| Concept: Circle        | 'Is that shape a circle? How do you know?' 'Name one circle in this room.' | Child can identify circles and explain no corners   |
| Concept: Centre        | 'Where did you put your first dot?' 'Is your centre in the middle?'        | Centre dot is roughly equidistant from all edges    |
| Concept: Pattern       | 'Tell me your colour pattern.' 'What comes next?'                          | Child can state and continue a repeating pattern    |
| Understanding: Mandala | 'What does Mandala mean?' 'How is this like a Rangoli?'                    | Links 'circle in Sanskrit' and Indian art tradition |
| Process: Creation      | 'What step are you on?' 'Which circle will you trace next?'                | Following steps sequentially; using tools correctly |

### Differentiation Strategies

| Learner Group          | Adaptations                                                                                                                                                                                                                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ■■ Needing Support     | <ul style="list-style-type: none"> <li>• Pre-draw the centre dot for the student</li> <li>• Provide pre-drawn concentric circles to work from</li> <li>• Reduce to 4 petals instead of 8</li> <li>• Use only 2 colours for colour pattern</li> <li>• Pair with a peer buddy for tracing steps</li> </ul>                                                  |
| ■ At Expected Level    | <ul style="list-style-type: none"> <li>• Follow all 6 steps independently with guidance</li> <li>• Create an ABAB or ABC repeating colour pattern</li> <li>• Draw 6–8 repeating petals around the circle</li> <li>• Add dots between petals</li> </ul>                                                                                                    |
| ■■ Extension Challenge | <ul style="list-style-type: none"> <li>• Add a 4th outer ring with a different repeating pattern</li> <li>• Create an ABCABC colour pattern</li> <li>• Draw two layers of petals (inner and outer)</li> <li>• Write the maths vocabulary words on the back of their mandala</li> <li>• Design their own Rangoli border pattern around the edge</li> </ul> |

## TEACHER NOTES & CULTURAL CONTEXT

### Cultural Context: Why Mandalas?

The mandala is one of the most mathematically rich art forms in Indian culture. By connecting geometry to familiar cultural traditions, we make maths *meaningful and joyful* for young learners — particularly those from Indian backgrounds who may recognise these patterns from home festivals, temple visits, or family traditions.

- **Rangoli** — Drawn at the entrance of homes, especially during Diwali, Pongal, and Onam. Geometric circles, dots, and lines form intricate repeating patterns.
- **Kolam (Tamil Nadu & South India)** — A traditional floor art using rice flour dots connected by curved lines. Highly mathematical — based on dot grids and symmetry.
- **Madhubani (Bihar)** — Folk painting tradition featuring highly geometric circular motifs, animals, and nature, painted using fingers and twigs.
- **Temple Ceilings** — The ceiling mandala (often called 'Padmanabha' or lotus mandala) is found in South Indian temples. Perfect rotational symmetry with 8 or 16 petals.
- **The Sanskrit word 'Mandala' (मण्डल)** literally means 'circle' or 'disc'. In Hindu and Buddhist traditions, mandalas represent the universe and wholeness.

### Classroom Management Tips

- ◆ **Materials setup:** Pre-arrange materials in small groups before class. Place coins and bowls in a small tray so students can select sizes.
- ◆ **Pacing:** Give a 2-minute warning before each transition. Use a gentle sound cue (bell or clap pattern) to signal 'pencils down, eyes on me'.
- ◆ **Noise level:** Create-Time is naturally buzzy — set expectation: 'We can whisper-talk while we create.' Use a noise-o-meter visual if helpful.
- ◆ **Tracing difficulty:** Some students will struggle to keep round objects still. Allow them to ask a partner to hold the object while they trace.
- ◆ **Display:** All mandalas deserve to be displayed! Consider a class 'Mandala Gallery' wall — this validates every child's creative-mathematical effort.

### Connecting to Next Session: Spirographs

The next session extends this lesson into **Spirographs** — overlapping circles that create intricate geometric flower patterns. The concepts of circle, centre, and pattern are the direct mathematical foundations for understanding spirograph geometry. Preview this with students: 'What happens if you put many circles together, all touching the same centre point?' This creates productive curiosity and motivates return to the next session.

### 3 Recall Questions (End of Session)

| # | Question                                                           | Expected Answer |
|---|--------------------------------------------------------------------|-----------------|
| 1 | What shape is a Mandala?                                           | CIRCLE! ■       |
| 2 | What is the middle of a circle called?                             | CENTRE! ■       |
| 3 | When a shape or colour repeats again and again, it is called a...? | PATTERN! ■      |

## STUDENT WORKSHEET – MY MANDALA JOURNAL

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Class: \_\_\_\_\_

### Part A: What I Learned Today

|                            |                                       |
|----------------------------|---------------------------------------|
| A circle is... (draw one!) | The centre of a circle is...          |
| A pattern is...            | Mandala means... (language: Sanskrit) |

### Part B: My Colour Pattern

Write or draw your colour pattern below. Circle what type it is:

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| Colour 1 | Colour 2 | Colour 1 | Colour 2 | Colour 1 | Colour 2 | Colour 1 | Colour 2 |
|          |          |          |          |          |          |          |          |

My pattern type (circle one): **ABAB** / **ABCABC** / **AABBAABB** / My own: \_\_\_\_\_

### Part C: My Mandala Steps Checklist

- ☐ I put a centre dot in the middle of my paper.
- ☐ I traced a small circle (coin/bangle) around it.
- ☐ I traced a bigger circle around that.
- ☐ I drew repeating petals between the circles.
- ☐ I added dots or lines between the petals.
- ☐ I coloured my mandala with a repeating pattern.

### Part D: My Reflection

The most interesting thing I learned today was:

\_\_\_\_\_

The maths I used while creating my mandala was:

\_\_\_\_\_

How did you feel about today's session? (circle one)

■ Needs more practice

■ I got it with help

■ I got it!

■ I can teach a friend!